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Paper 1: Nipocalimab Dose Selection in Generalized Myasthenia Gravis.

Bibliographic Information

- **Title**: Nipocalimab Dose Selection in Generalized Myasthenia Gravis.
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Quick Take

This paper presents a mechanistic population PK/RO/IgG/MG-ADL model for nipocalimab, an anti-FcRn monoclonal antibody, integrating data from five phase 1 studies and one phase 2 study in generalized myasthenia gravis (gMG). Model-based simulations identified 15 mg/kg Q2W (after a 30 mg/kg loading dose) as the optimal regimen for the pivotal phase 3 trial, achieving ~70% IgG reduction and clinically meaningful MG-ADL improvement. Pharmacometricians and clinical developers in autoimmune diseases will find this a strong example of model-informed drug development for dose selection.

Executive Summary

This study integrates pharmacokinetic (PK), FcRn receptor occupancy (RO), total serum IgG, and efficacy (MG-ADL) data from 228 participants (160 healthy, 68 gMG) across six clinical studies to develop a mechanistic population model for nipocalimab. The model comprises a two-compartment PK model with target-mediated disposition (QSS approximation), an indirect-response model for IgG with FcRn-dependent recycling, and a placebo-corrected drug effect model linking relative IgG change from baseline to MG-ADL improvement via an effect compartment. Key findings include: nipocalimab exhibits nonlinear, FcRn-mediated clearance with no accumulation upon Q2W dosing; PK and IgG reduction are similar between healthy and gMG participants; each 10% reduction in IgG yields a 0.22-point MG-ADL improvement at median baseline (MG-ADL0=7). Simulations predicted that 15 mg/kg Q2W (with a 30 mg/kg loading dose) achieves the target of 70% average IgG reduction and provides a clinically meaningful MG-ADL improvement, leading to its selection for the phase 3 Vivacity-MG3 study, which subsequently confirmed efficacy. This work exemplifies model-informed drug development for dose selection in autoimmune diseases.

Scientific Context & Motivation

Generalized myasthenia gravis (gMG) is an autoantibody-mediated neuromuscular disease with an unmet need for safer, more effective therapies. Nipocalimab, a fully human IgG1 monoclonal antibody, blocks the neonatal Fc receptor (FcRn), accelerating IgG catabolism and reducing pathogenic autoantibodies. While phase 1 and phase 2 studies showed promising efficacy, the optimal dose regimen for phase 3 was unknown. This study addresses the gap by developing a

quantitative mechanistic framework that integrates PK, target engagement (RO), biomarker (IgG), and clinical efficacy (MG-ADL) to guide dose selection. It also provides a template for model-based dose selection for other anti-FcRn therapies and autoimmune indications.

⚡ Methodological Snapshot

A mechanistic population model was developed using nonlinear mixed-effects modeling. The PK component used a two-compartment model with parallel linear and target-mediated (FcRn) clearance, employing a quasi-steady-state approximation for receptor binding. An indirect-response model described total serum IgG with FcRn-dependent recycling. The efficacy component linked relative IgG change from baseline to MG-ADL improvement via a linear relationship with an effect compartment. Model-based simulations with 10,000 virtual participants were used to compare dosing scenarios and select the phase 3 regimen.

⊠ Structural Model Breakdown

The PK model is a two-compartment model with central (Vc) and peripheral (Vp) volumes, intercompartmental clearance (Q), and linear clearance (CL). Target-mediated disposition is modeled via binding to FcRn: free drug binds to free receptor (FcRn) with second-order rate kon, the complex dissociates (koff) or is internalized (kint). The quasi-steady-state approximation uses Kss = (koff + kint)/kon. FcRn is synthesized at zero-order rate ksyn and degrades at first-order rate kdeg. The maximum fraction of FcRn available for binding is FRmax. The IgG model is an indirect-response model: IgG is synthesized at zero-order rate $ksyn_{IgG}$ and eliminated via first-order degradation ($kdeg_{IgG}$) with a recycling component ($krec_{IgG}$) proportional to free FcRn fraction. The MG-ADL model is an additive model: baseline MG-ADL0 plus placebo effect (initial drop and linear slope) plus drug effect driven by relative IgG CFB through an effect compartment (ke0). The drug effect slope is modulated by MG-ADL0 via an exponent.

Detailed Methodological Analysis

Modeling Approach

Nonlinear mixed-effects modeling (population approach) using a mechanistic target-mediated drug disposition model for PK, an indirect-response model for IgG, and a placebo-corrected drug effect model for MG-ADL. The PK/RO/IgG model was developed first, and individual parameters were used to predict IgG CFB as input to the IgG/MG-ADL model (sequential approach). Allometric scaling was applied to PK parameters (CL, Vc, Q, Vp) with exponents 0.75 and 1.0 for clearances and volumes, respectively. Software: likely NONMEM (not explicitly stated, but typical for such analyses).

Data Sources

Data from five phase 1 studies in healthy participants (N=160) and one phase 2 study (Vivacity-MG) in gMG participants (N=68). PK samples were collected intensively; RO was measured in some phase 1 studies; total serum IgG was measured in all studies; MG-ADL was collected in

the phase 2 study. Dosing regimens ranged from 0.3 mg/kg single dose to 60 mg/kg Q2W. Body weight, age, sex, race, renal/hepatic function, ADA status, and baseline IgG were recorded.

Estimation Methods
First-order conditional estimation with interaction (FOCE-I) or similar likelihood-based method (not explicitly stated). The M3 method was used to handle PK concentrations below the limit of quantification. Interindividual variability was modeled with exponential errors (except additive for IgG/MG-ADL model). Residual variability was modeled with additive+proportional for PK and RO, proportional for IgG, and additive for MG-ADL.

Model Evaluation
Visual predictive checks (VPCs) for PK, RO, IgG, and MG-ADL; goodness-of-fit plots; assessment of parameter precision via relative standard errors (RSE). No bootstrap or cross-validation was reported.

Covariate Analysis
Body weight was included as an allometric covariate on PK parameters. Baseline MG-ADL (MG-ADL0) was found to influence both placebo effect and drug effect slope. Study-specific differences (e.g., FRmax in study EDI1002, IgG0 in MOM-M281-004) were accounted for. No other covariates (age, sex, race, renal/hepatic function, ADA) were retained in the final model.[1]

Statistical Rigor Assessment
The model was estimated with adequate precision for PK/RO/IgG parameters (RSE < 23% for fixed effects, < 19% for random effects). However, the IgG/MG-ADL model parameters had high uncertainty (RSE 32-47%) due to small sample size (n=67) and high variability in MG-ADL scores (residual SD = 1.5 points). The sequential modeling approach may underestimate uncertainty in the efficacy parameters. No sensitivity analyses or bootstrap were reported. The M3 method for BLOQ data is appropriate. The use of VPCs is standard, but no external validation was performed. Overall, the statistical rigor is adequate for the primary goal of dose selection, but the efficacy model should be interpreted with caution.

□ Key Findings
Nipocalimab PK was best described by a two-compartment model with parallel linear and target-mediated (FcRn) clearance. The linear clearance was 0.655 L/day (for 75 kg), and the quasi-steady-state constant Kss was 6.05 μg/mL. FcRn baseline (FcRn0) was 143 nM with a maximal fraction available for binding (FRmax) of 0.947. IgG recycling was modeled with an indirect-response model; at full FcRn blockade, IgG half-life decreased from 19.3 to 3.19 days, yielding a maximum IgG reduction of 83.6%. The IgG/MG-ADL relationship was linear: each 10% reduction in relative IgG CFB produced a 0.22-point decrease in MG-ADL pcCFB at MG-ADL0=7, with a small delay (effect compartment half-life 1.17 days). Simulations showed that 15 mg/kg Q2W (after 30 mg/

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kg loading) achieved the target of 70% average IgG reduction and provided a 1.94-point MG-ADL improvement, while higher doses (30 mg/kg Q2W) or less frequent dosing (30 mg/kg Q4W) offered no meaningful advantage. The selected regimen was later confirmed in the phase 3 Vivacity-MG3 study.

Clinical & Regulatory Implications

The model-informed dose selection identified 15 mg/kg IV Q2W (with a 30 mg/kg loading dose) as the optimal regimen for gMG, achieving a target 70% average IgG reduction and clinically meaningful MG-ADL improvement. This regimen was subsequently confirmed in the phase 3 Vivacity-MG3 study (1.45-point MG-ADL improvement, p=0.002). The loading dose provides faster onset of effect. The model also suggests that higher doses (30 mg/kg Q2W) or less frequent dosing (30 mg/kg Q4W) are not advantageous. For regulatory purposes, this analysis supports the dose rationale and may inform label recommendations. The framework can be extended to other autoimmune indications where FcRn blockade is being explored.

Strengths & Limitations

Strengths

- Comprehensive integration of PK, RO, IgG, and efficacy data from multiple studies and populations (healthy and gMG).
- Mechanistic model with target-mediated drug disposition and FcRn-dependent IgG recycling, grounded in biology.
- Use of quasi-steady-state approximation to reduce model complexity while preserving mechanistic insight.
- Model-based simulations with realistic covariate distributions (body weight, MG-ADL0) to support dose selection.
- Prospective validation: the selected dose was later confirmed in the phase 3 Vivacity-MG3 study, demonstrating the predictive value of the model.
- Adequate parameter precision for PK/RO/IgG components (RSE < 23%).

Limitations (Acknowledged by Authors)

- High uncertainty (RSE 32-47%) in IgG/MG-ADL model parameters due to limited sample size (n=67) and high variability in MG-ADL scores.
- RO data were not measured in the phase 2 gMG study; RO was inferred from the PK/IgG model.
- kdeg (FcRn degradation rate) was fixed to 1.3 day^{-1} to enable convergence, based on literature.
- Sequential modeling approach (PK/RO/IgG then IgG/MG-ADL) may propagate uncertainty.
- The placebo effect model was derived from a single phase 2 study and may not generalize.[2]

[2] **UNSUPPORTED** — original: "The placebo effect model was derived from a single phase 2 study and may not generalize." → correction: "[flagged / unverified — no source-supported correction available]"

Limitations (Expert Review)

- No external validation of the IgG/MG-ADL model beyond the phase 2 data; the model was not validated against independent datasets.
- The linear relationship between IgG reduction and MG-ADL improvement may not hold at extremes or for long-term treatment.[3]
- The effect compartment is empirical; the biological delay between IgG reduction and clinical improvement is not mechanistically explained.
- Body weight allometric scaling used fixed exponents (0.75 and 1.0) without testing alternative relationships.
- The model assumes 1:1 stoichiometry and rapid equilibrium for FcRn binding, which may oversimplify in vivo conditions.
- Potential bias from pooling studies with different assays (ELISA vs ECLIA) for PK and RO measurements, though residual error was modeled separately.

Generalizability

The model is specific to nipocalimab and its FcRn mechanism, but the framework (PK/RO/biomarker/efficacy) is highly transferable to other anti-FcRn monoclonal antibodies and IgG-mediated autoimmune diseases. The dose-selection methodology (target IgG reduction linked to clinical outcome) can be adapted to other indications. However, the quantitative relationships (e.g., slope between IgG and MG-ADL) are disease- and drug-specific and should not be extrapolated without further data.

Key Equations

Target-mediated drug disposition (QSS approximation)

$$\frac{dA_c}{dt} = -\left(\frac{CL}{V_c} + \frac{Q}{V_c}\right)A_c + \frac{Q}{V_p}A_p - k_{on} \cdot A_c \cdot R_{free} + k_{off} \cdot C_{complex} \cdot V_c$$

Central compartment mass balance for free nipocalimab, including linear clearance, distribution, and binding to FcRn. The quasi-steady-state approximation assumes rapid equilibrium and uses Kss = (koff + kint)/kon.

FcRn receptor dynamics

$$\frac{dR_{free}}{dt} = k_{syn} - k_{deg} \cdot R_{free} - k_{on} \cdot C_{free} \cdot R_{free} + k_{off} \cdot C_{complex}$$

Free FcRn receptor concentration changes due to synthesis, degradation, and binding to nipocalimab. ksyn is derived from baseline conditions: ksyn = kdeg * FcRn0 * FRmax.

IgG indirect response model

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$$\frac{dIgG}{dt} = k_{syn_IgG} - \left(k_{deg_IgG} - k_{rec_IgG} \cdot \frac{R_{free}}{FcRn_0}\right) \cdot IgG$$

Total serum IgG dynamics with FcRn-dependent recycling. $ksyn_{IgG}$ = ($kdeg_{IgG}$ - $krec_{IgG}$) * IgG0, and $krec_{IgG}$ = (IgK/(1+IgK)) * $kdeg_{IgG}$, where IgK is estimated.

MG-ADL drug effect model

$$MG\text{-}ADL(t) = MG\text{-}ADL_0 + \Delta MG\text{-}ADL_P(t) + S_{IgG} \cdot \left(\frac{MG\text{-}ADL_0}{7}\right)^{E_{IgG}} \cdot A_e(t)$$

Total MG-ADL score as sum of baseline, placebo effect, and drug effect. Ae is the effect compartment concentration driven by relative IgG CFB, with ke0 = 0.414 day^{-1}.

Figures & Tables

- **Figure 1**: Schematic of the nipocalimab PK/RO/IgG/MG-ADL model showing compartments for central/peripheral PK, FcRn binding, IgG dynamics, and MG-ADL effect.
  - *Significance*: Provides a visual overview of the integrated mechanistic model, illustrating the connections between drug concentration, receptor occupancy, IgG reduction, and clinical response.
- **Figure 2**: Prediction-corrected visual predictive checks (VPCs) for nipocalimab serum concentrations, FcRn occupancy, total serum IgG percentage change from baseline, and MG-ADL absolute change.
  - *Significance*: Demonstrates the model's ability to capture the central tendency and variability of the observed data across all endpoints, supporting model adequacy.
- **Figure 3**: Simulated time course of total serum IgG percentage change from baseline for different dosing regimens: (A) 15 vs 30 mg/kg Q2W, (B) 15 mg/kg Q2W vs 30 mg/kg Q4W, (C) with vs without 30 mg/kg loading dose.
  - *Significance*: Key simulation results that directly informed dose selection, showing that 15 mg/kg Q2W achieves the target 70% IgG reduction with minimal additional benefit from higher doses and that Q2W is superior to Q4W.
- **Figure 4**: Typical profiles of free nipocalimab concentration, FcRn inhibition, total serum IgG change from baseline, and MG-ADL change from baseline for 7.5, 15, and 30 mg/kg Q2W regimens (with loading doses).
  - *Significance*: Illustrates the predicted time course of PK, target engagement, biomarker, and clinical response for the considered doses, aiding in understanding the dose-response relationship.
- **Table 1**: Summary of the six clinical studies included in the analysis, with populations, dosing regimens, and sampling schedules for PK, RO, IgG, and MG-ADL.
  - *Significance*: Provides the data foundation for the model, showing the breadth of information integrated (healthy and gMG participants, multiple dose levels).
- **Table 2**: Baseline demographics and clinical characteristics of the analysis population (healthy vs gMG).

- *Significance*: Highlights differences between populations (e.g., age, body weight, IgG levels) that were accounted for in the model, and supports generalizability.
- **Table 3**: Final parameter estimates for the PK/RO/IgG/MG-ADL model, including fixed effects, interindividual variability, and residual variability.
  - *Significance*: Provides the quantitative backbone of the model, allowing readers to reproduce simulations and assess parameter precision.

Code & Reproducibility Assessment

No code or model files were provided. Data sharing is available through the Johnson & Johnson YODA project (https://yoda.yale.edu) upon request, but no analysis code is publicly accessible.

Supplementary Materials

Supplementary materials include Text S1 (MG-ADL description), Text S2 (bioanalytical methods for PK, RO, and IgG), Text S3 (software used), Text S4 (full model equations), and Figure S1 (individual model fits for representative participants). These provide additional methodological detail and support reproducibility.

Future Directions

Future work could include: (1) external validation of the IgG/MG-ADL model using phase 3 data; (2) exploration of alternative dosing regimens (e.g., subcutaneous administration) using the model; (3) extension to other anti-FcRn therapies and autoimmune indications; (4) incorporation of longitudinal autoantibody subtype data (e.g., anti-AChR) to refine the efficacy relationship; (5) development of a more mechanistic link between IgG reduction and clinical response, potentially including disease severity biomarkers.

Expert Commentary

This is a textbook example of model-informed drug development (MIDD) for dose selection in a rare autoimmune disease. The authors skillfully combine a target-mediated drug disposition model with an indirect-response model for IgG and a pharmacometric model for clinical efficacy, all within a nonlinear mixed-effects framework. The decision to use a quasi-steady-state approximation for FcRn binding is pragmatic and well-justified. The key strength is the prospective confirmation of the selected dose in the phase 3 study, which validates the model's predictive utility. However, the high uncertainty in the efficacy model parameters highlights the challenges of modeling clinical endpoints with high variability and small sample sizes. The sequential modeling approach, while practical, may underestimate uncertainty; a joint estimation would be more rigorous but computationally intensive. Overall, this paper sets a high standard for dose selection in biologics development and will be a reference for future anti-FcRn programs.

Bottom Line

For practicing pharmacometricians, this paper demonstrates a rigorous model-informed approach to dose selection for a monoclonal antibody targeting FcRn. The integration of PK, target occupancy, biomarker (IgG), and clinical efficacy into a single mechanistic model allowed identification of the lowest effective dose (15 mg/kg Q2W with a 30 mg/kg loading dose) that achieves a target 70% IgG reduction, which was later confirmed in a phase 3 trial. The work underscores the value of quantitative systems pharmacology in guiding pivotal trial design and provides a template for similar programs in autoimmune diseases.

Fact-check corrections

Figures

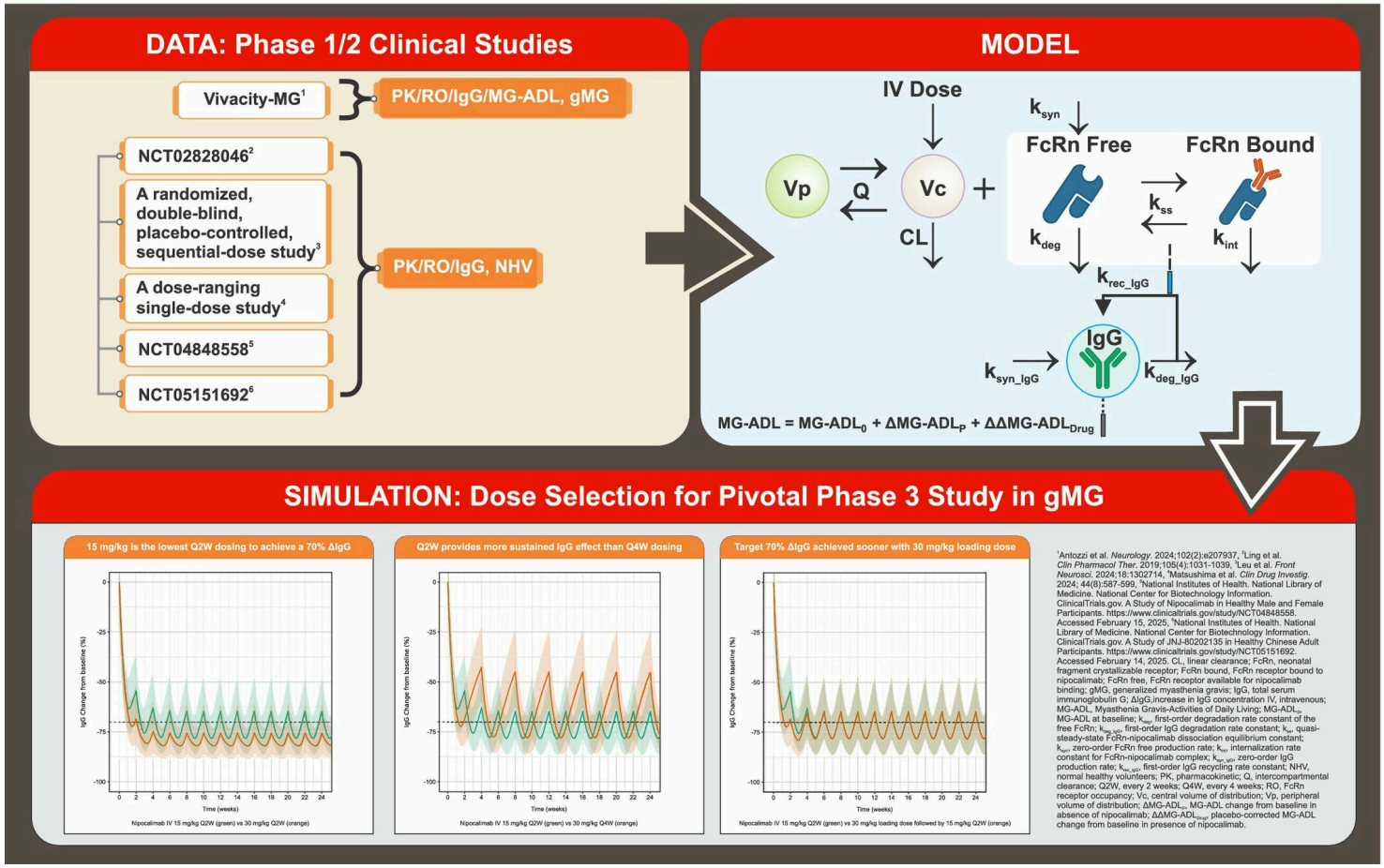


Figure 1: Figure 1

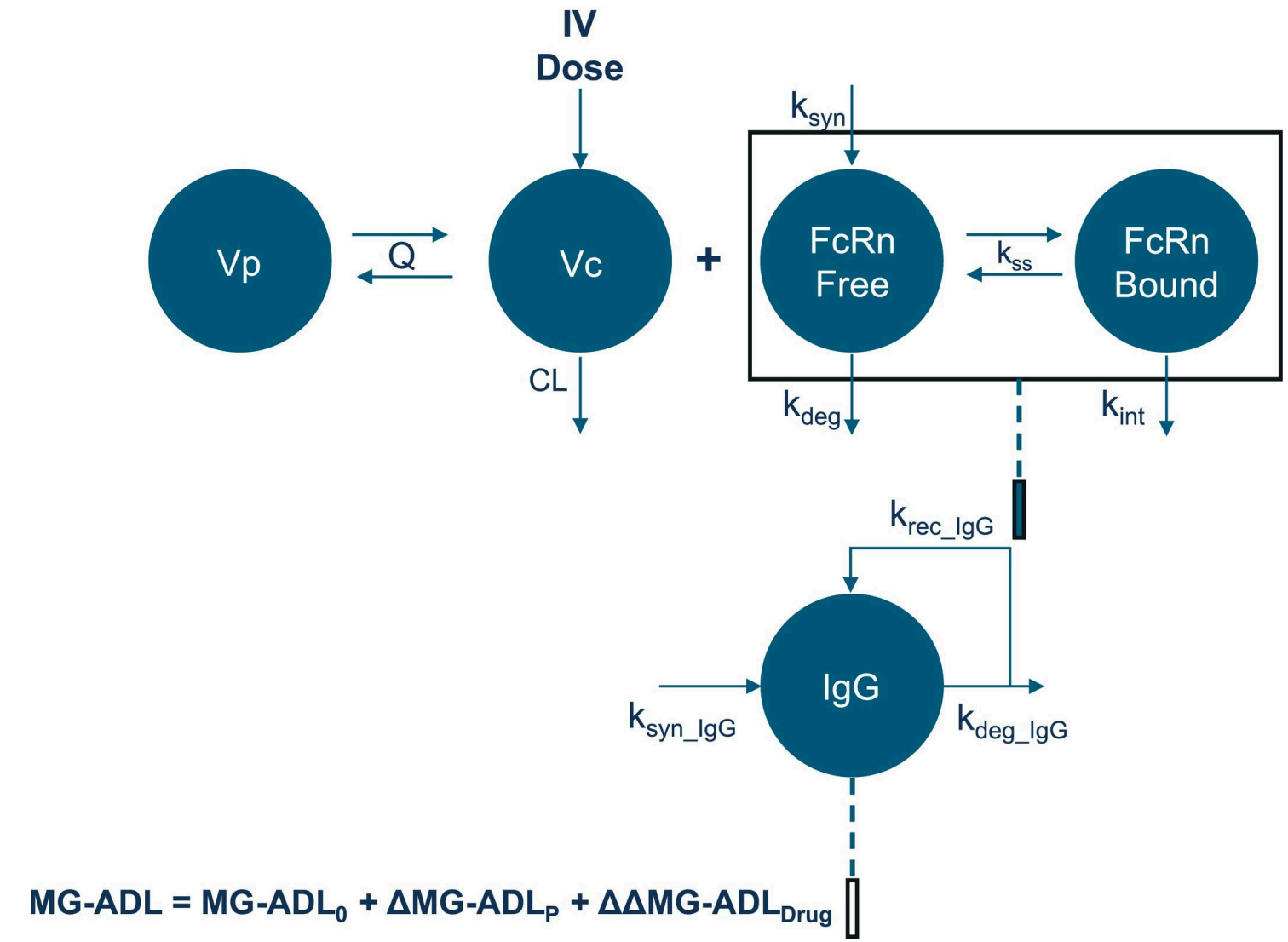


Figure 2: Schematic of the nipocalimab PK/RO/IgG/MG-ADL model. After IV administration, free nipocalimab is distributed into the volume of the central compartment (Vc). No

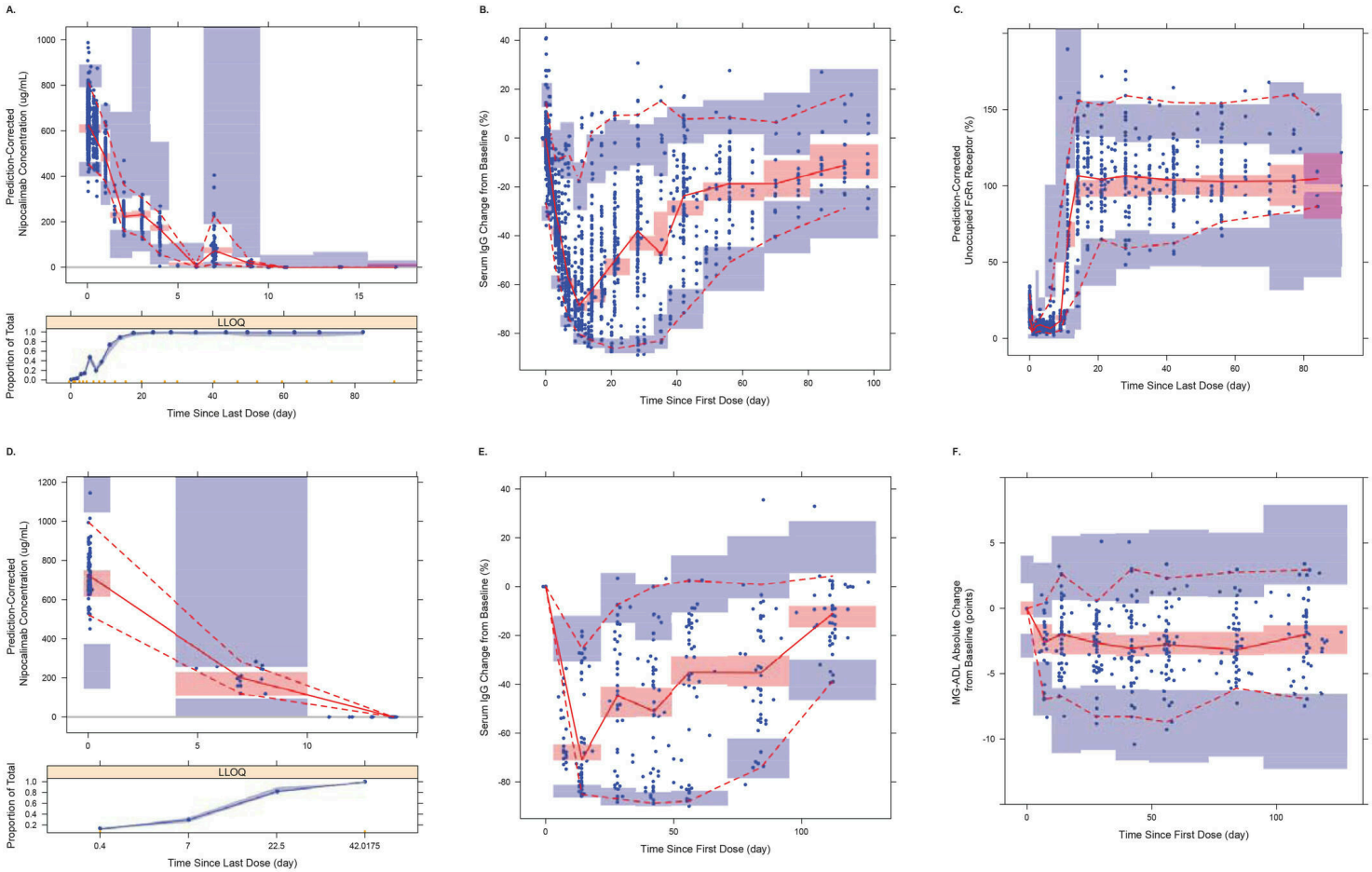


Figure 3: Prediction-corrected visual predictive check of nipocalimab serum concentrations, FcRn occupancy, total serum IgG percentage change from baseline, and MG-ADL abs

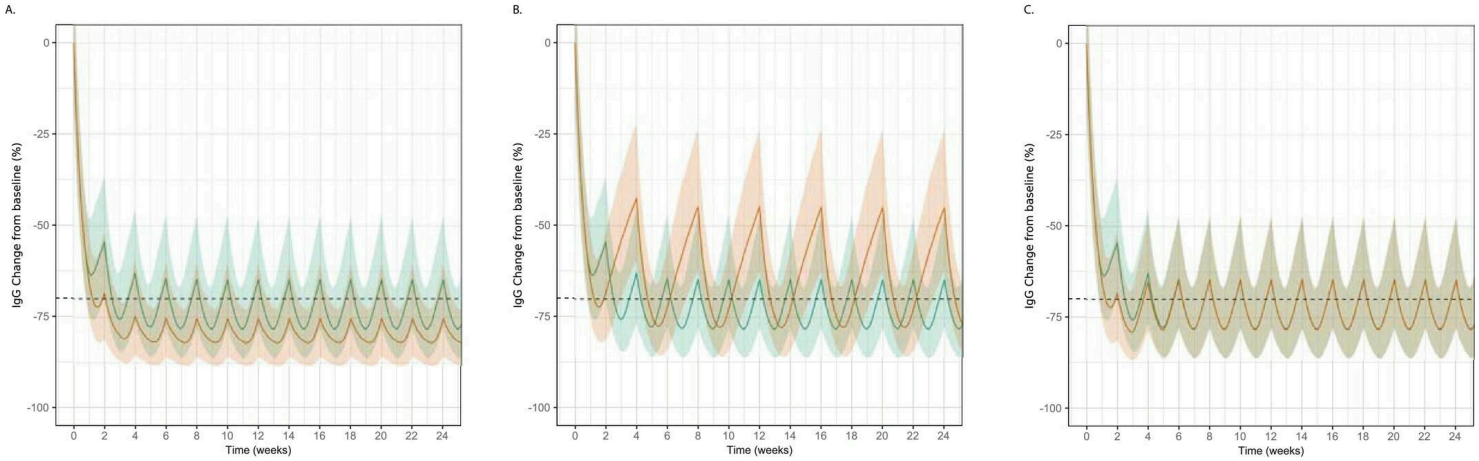


Figure 4: Time course of total serum IgG percentage change from baseline after nipocalimab IV 15 mg/kg Q2W (green) versus 30 mg/kg Q2W (Panel A, orange), 30 mg/kg Q4W (Pan

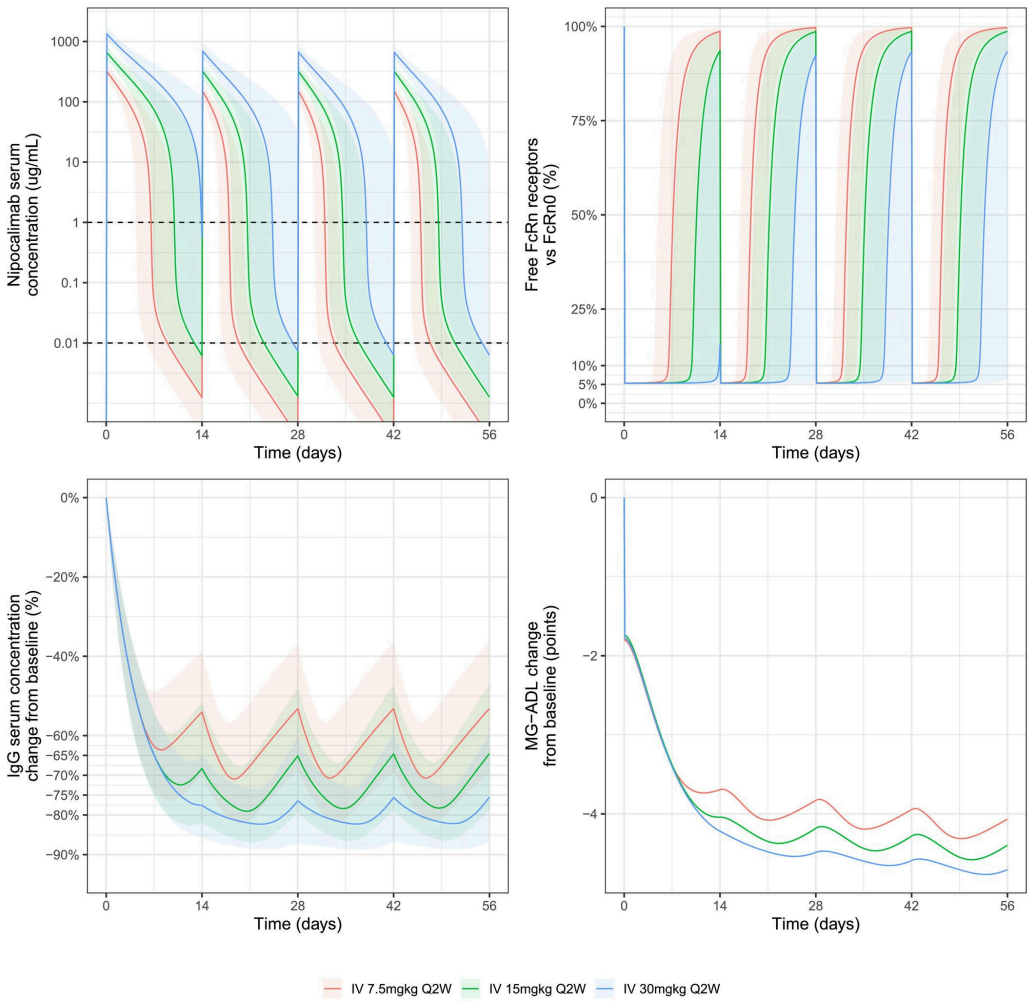


Figure 5: Nipocalimab free serum concentration (top left), FcRn inhibition (top right), total serum IgG change from baseline (bottom left), and MG-ADL change from baseline